

JOSHUA NWACHINEMERE

AI Engineer | Applied AI, ML Systems, Automation

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PROFILE

AI Engineer and VolyxAI founder building applied AI, backend services, and automation systems with Python, JavaScript, and TypeScript. Experience includes a multimodal macOS assistant, retrieval workflows, real-time applications, ML evaluation, API integrations, and human-in-the-loop automation.

CORE SKILLS

- Languages: Python, JavaScript, TypeScript, SQL
- AI / ML: Azure AI Foundry, XGBoost, scikit-learn, pandas, Ollama, LangChain, Chroma, RAG, model evaluation
- Backend & Automation: FastAPI, Express, Socket.IO, n8n, REST APIs, webhooks, SQLAlchemy, SQLite
- Infrastructure: Docker, GitHub Actions, Linux, Microsoft Azure, Electron
- Security: Suricata, Snort, VirusTotal API, OSINT, DNS analysis, secure credential management

PROFESSIONAL EXPERIENCE

Founder & AI Automation Engineer | Nov 2025 - Present

VolyxAI

- Built and tested controlled workflow prototypes connecting conversational intake to schema validation, calendar scheduling, email confirmation, and approval-based handoffs.
- Integrated language models, n8n, webhooks, Vapi, ElevenLabs, Google Calendar, and email services in controlled test environments.
- Designed deterministic validation, bounded retries, idempotency controls, and human approval for consequential actions.

Key tools: Python, n8n, TypeScript, REST APIs, webhooks, Vapi, ElevenLabs, Azure AI Foundry

Independent Software & Automation Developer | Jan 2021 - Present

Self-Employed / Independent Projects

- Built Python automation and backend utilities for data processing, Telegram bots, API integrations, and operational tooling.
- Developed stream-oriented utilities for processing large text datasets without loading complete files into memory.
- Created asynchronous integrations for wallet analysis, malware scanning, user tracking, pagination, caching, and admin reporting.

Key tools: Python, Async I/O, Telegram Bot API, aiohttp, SQL, Docker, REST APIs

TECHNICAL TRAINING

Threat Detection & Response Capstone | Capstone IT Staffing (Remote) | Jul 2025 - Aug 2025

- Deployed Suricata on Ubuntu, wrote custom Snort rules, and automated alert enrichment with Python and public OSINT sources.
- Built a DNS intelligence CLI with asynchronous resolution, WHOIS lookups, trace output, and JSON export.

SELECTED PROJECTS

Volyx Lens | Privacy-First Context-Aware AI Assistant

JavaScript, Electron, Swift, Azure AI Foundry, Multimodal APIs

- Developed and substantially extended a macOS assistant that combines explicitly selected screen context, microphone input, and meeting audio with user-configured AI providers.
- Implemented local OCR, bounded screenshot memory, context ranking, provider routing, and consent before larger image uploads.
- Hardened Electron boundaries with sandboxing, context isolation, restricted IPC, automated tests, secret scanning, and release checks.
- Source: <https://github.com/dk3yyyy/volyx-lens>

Football Predictor | Experimental ML Pipeline

Python, XGBoost, scikit-learn, FastAPI, Streamlit, SQLAlchemy

- Built data scrapers, temporal feature engineering, XGBoost outcome and Poisson goal models, a FastAPI service, and a Streamlit dashboard.
- Evaluated with rolling-origin testing across 1,140 Premier League matches: 53.77% outcome accuracy versus a 56.70% bookmaker-implied benchmark, showing signal but no practical betting advantage.
- Archived evidence: https://github.com/dk3yyyy/football_predictor/tree/repair/football-predictor-hardening

Noughtline | Real-Time Multiplayer System

React, Express, Socket.IO, SQLite, Paystack

- Built signed guest sessions, server-authoritative moves, reconnect and forfeit handling, transactional match settlement, and an append-only currency ledger.
- Added idempotent Paystack crediting plus automated API, economy, room-state, and two-client Socket.IO tests.
- Live: <https://noughtline.onrender.com> | Source: <https://github.com/dk3yyyy/Noughtline>

VirusTotal Telegram Bot | Security API Integration

Python, Pyrogram, aiohttp, VirusTotal API

- Built asynchronous file, URL, and hash scanning with local SHA-256 lookup, VirusTotal large-file upload handling, bounded retry backoff, and short-lived report caching.
- Source: <https://github.com/dk3yyyy/VirusTotal-Telegram-Bot>

CERTIFICATIONS

- Google AI Professional Certificate | Coursera
- Google Cybersecurity Professional Certificate | Coursera

EDUCATION

Bachelor of Technology (BTech), Mathematics | 2016 - 2021

Federal University of Technology, Owerri (FUTO)